Full-State Reconstruction of Fusion Plasmas from Sparse Diagnostics Using Continuous-Time Latent Dynamics Models

Mid-Review Presentation

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[FILL IN: Submission Date] – [FILL IN: Semester]

Problem Statement & Motivation

Problem. Magnetic fusion plasmas are only partially observable: reactor-relevant quantities (density, temperature, electric field, flow) live inside the hot plasma where sensors cannot survive; available diagnostics are sparse, line-integrated, or boundary-only.

Given only a handful of local sensor time-series, **reconstruct the full 2D spatial state** of the plasma (multiple coupled physical fields) at any requested time.

Why deep learning? The sensor-to-field map is a nonlinear, high-dimensional inverse problem. Physics-based inversions are slow and ill-posed; a learned model can capture the coupling between measured and unmeasured quantities from simulation/experiment archives and run in milliseconds – fast enough for real-time use.

Who is affected?

- Tokamak operators / control systems (real-time state estimation, disruption avoidance)
- Diagnostics engineers (virtual diagnostics / "digital twin")
- Fusion pilot plants that must infer state when magnetic diagnostics fail in neutron-rich reactors

Background / Problem Context

- Tokamak diagnostics are **heterogeneous**: local point probes (Langmuir, Thomson-scattering points), line-integrated signals (interferometer, bolometer, SXR), and boundary magnetic coils – sampling rates differ and channels drop out.
- Reconstructing state from such measurements is an **ill-posed inverse problem** (tomography, equilibrium reconstruction, profile inversion).
- Existing deep-learning approaches fall into three families:
  1. Static / snapshot mapping
  2. LSTM / Transformer temporal encoders
  3. Physics-informed solvers (equilibrium lane)
- **Continuous-time latent dynamics** (Neural ODE, LTC, CfC) is a mature paradigm for irregularly-sampled, dropout-prone time series with documented robustness to missing data – but it has rarely been applied to plasma reconstruction.

Literature Review – Overview

Prior work relevant to this project falls into five threads:

A. Full-state reconstruction from sparse sensors (closest prior art)

B. Real-device profile / parameter reconstruction from diagnostics

C. Tomography – 2D emissivity / state inversion from line-integrated measurements

D. Equilibrium / inverse reconstruction (saturated physics-informed lane)

E. Latent / missing-data representations on real plasma (adjacent methods)

We review each thread next, followed by a comparison table (required by the rubric) summarizing all 15 papers considered.

Literature Review – Threads A & B

(A) Full-state reconstruction from sparse sensors. The closest prior art (Reza & Faraji, 2026) reconstructs multiple coupled plasma fields from as few as three density sensors using an LSTM temporal encoder feeding a spatial decoder, building on the Shallow Recurrent Decoder (SHRED) framework originally proposed for sensing dynamical systems from sparse point sensors (Williams, Zahn & Kutz, 2023) and later shown to generalize across E×B plasma configurations (SHRED for E×B plasmas, 2025). All of this family assumes **regular, complete sampling** and uses **discrete-time** recurrence.

(B) Real-device profile / parameter reconstruction. On real tokamaks, deep networks reconstruct 1D kinetic profiles in real time – e.g. RTCAKENN on DIII-D reconstructs seven profiles in under 8 ms while training with diagnostic dropout for robustness (Shousha et al., 2023) – or map indirect line-integrated signals to profiles (COMPASS temporal super-resolution, 2026; Onion physics-informed line-integral network, 2025; PIDP-KAN on EAST, 2025), and a transformer-based extrinsic regressor reconstructs missing diagnostic channels on EAST with high confidence (TSER, 2025). These works motivate robustness to missing/indirect data but remain **1D, static, or discrete-time**.

Literature Review – Threads C, D & E

(C) Tomography. Deep networks have long been used for fast 2D emissivity inversion from line-integrated measurements: CNN-based bolometer tomography at JET (Ferreira et al., 2018), U-Net plus deep-unfolded SIRT for multispectral imaging on TCV (2025), and a time-resolved physics-informed network for transient emission events (TRT-PINN, 2025). These establish deep 2D inversion as fast and effective, but are largely **static, per-frame** reconstructions.

(D) Equilibrium / inverse reconstruction. Physics-informed networks jointly reconstruct magnetic equilibrium and kinetic profiles at JET with very high fidelity (Rutigliano et al., 2026), and an open benchmark (Fusion Equilibrium Challenge, 2026) evaluates flux-map reconstruction across DIII-D and MAST, showing that **cross-machine generalization remains an open problem** even in this mature, heavily physics-constrained lane.

(E) Latent / missing-data representations. Closer to our approach, VAE-based latent-state models learn low-dimensional representations of the AUG plasma state and forward-predict profile evolution (Kit et al., 2023), and a masked-autoencoder pretraining scheme unifies 88 diagnostics on HL-3 with strong masked-channel reconstruction (FusionMAE, 2025). Both demonstrate the value of latent representations and missing-data robustness, but neither uses **continuous-time dynamics** nor targets **sparse-sensor full-field reconstruction**.

Literature Review – Comparison Table (A: Sparse-Sensor Full-State Reconstruction)

Paper (Year)		Problem / Dataset	Method	Key Result	Limitation / Relevance
Reza & Faraji (arXiv:2607.04390)	2026	Full-domain reconstruction from few sensors; 2D PIC E×B sim, 256×256, virtual sensors	LSTM temporal encoder + spatial decoder	Multiple coupled fields from 3 density sensors, robust to placement	Discrete-time, regular sampling, single testbed. Our direct baseline – we replace LSTM with continuous-time CfC.
SHRED for E×B Plasmas (MLST 2025)	2025	ROM/reconstruction for Hall-thruster E×B plasmas; multi-geometry sims	SHRED = LSTM + shallow decoder	Full state (incl. unmeasured fields) from 3 sensors; beats DMD	Discrete-time, fixed sampling. Same architecture family as baseline.
Williams, Zahn & Kutz (SHRED, 2023)	2023	Sense full state of a dynamical system from few point sensors; e.g. sea surface, fluids	Time-lagged sensor embeddings → LSTM → shallow decoder	Foundational full-field reconstruction from sparse trajectories	Fluid/other domains, discrete-time. Foundational method our architecture extends.

Literature Review – Comparison Table (B: Real-Device Profile Reconstruction, 1/2)

Paper (Year)		Problem / Dataset	Method	Key Result	Limitation / Relevance
Shousha et al., RTCAKENN, Nucl. Fusion 64:026006	2023	Fast kinetic profile reconstruction for real-time control; DIII-D CAKE/EFIT archives	DNN encoder-decoder with diagnostic dropout	7 profiles in <8 ms; robust to missing TS/CER	1D, snapshot, discrete-time. Real-device analogue; robustness via dropout, not continuous time.
COMPASS Temporal Super-Resolution, PPCF (ae72c6)	2026	μs-resolution T_e, n_e profiles from fast radiation diagnostics; COMPASS SXR+AXUV	NN mapping line-integrated SXR/AXUV to radial profiles	$\sim$10% core / $\sim$20% edge error; tracks ELMs/sawteeth	1D radial, static mapping. Different target space.
Onion (physics-informed, line-integral), Nucl. Fusion (ade0ce)	2025	Fast 2D profile reconstruction from line-integral measurements; synthetic + EAST/HL-2A SXR	Physics-informed architecture + line-integral loss (PILF)	Consistent error reduction, synthetic + experimental	Geometry-specific, no dynamics. Shows physics losses help.

Literature Review – Comparison Table (B: Real-Device Profile Reconstruction, 2/2)

Paper (Year)		Problem / Dataset	Method	Key Result	Limitation / Relevance
Shen et al., PIDP-KAN, Plasma Sci. Technol. (adca8e)	2025	Real-time density profile from polarimeter–interferometer; EAST POINT diagnostics	Kolmogorov–Arnold Network (spline activations)	$<5\%$ error, ~ 2 ms	1D profile, single diagnostic. Alternative NN backbone for profile inversion.
TSER for Missing T_e (EAST), Nucl. Fusion (addb5f)	2025	Reconstruct missing diagnostic channels; EAST discharge signals	Transformer time-series extrinsic regression	$>95.9\%$ confidence within 2σ	Reconstructs a signal, not a spatial field. Motivates our missing-data robustness claim.

Literature Review – Comparison Table (C: Tomography)

Paper (Year)		Problem / Dataset	Method	Key Result	Limitation / Relevance
Ferreira, Carvalho & Fernandes, Fusion Sci. Technol. 74(1-2):47	2018	Real-time bolometer tomography at JET; $\sim$10k samples	CNN / deconvolutional network	Accurate 196$\times$115 reconstructions, orders of magnitude faster than iterative methods	Static, regular inputs. Earliest deep reconstruction in fusion; precedent for image inversion.
ML-Enhanced Tomography for Multispectral Imaging (TCV), PPCF (ada856)	2025	Fast poloidal emissivity reconstruction; TCV MANTIS camera, synthetic+exp	Informed U-Net + deep-unfolded SIRT	Outperforms iterative SIRT; transferable across geometry	2D emissivity only, fixed geometry. Deep 2D inversion; transferability idea.
TRT-PINN, Nucl. Fusion (adb3bc)	2025	Time-resolved emission tomography of transients (ELM/MARFE); JET bolometer	Time-resolved physics-informed NN (diffusion constraint)	Good reconstruction with super-resolution, self-modelling	Training cost, tunable weights. Links time to inversion via a different mechanism than latent dynamics.

Literature Review – Comparison Table (D: Equilibrium / Inverse Reconstruction)

Paper (Year)		Problem / Dataset	Method	Key Result	Limitation / Relevance
Rutigliano et al., Nucl. Fusion 66:046006	2026	Equilibrium + kinetic profile inverse problem; JET + TokaLab synthetic	PINN with multi-diagnostic + MHD constraints	R^2 >99% synthetic; ablation reconstruction >97% Pearson	Equilibrium scope, heavy physics modeling, no dynamics. Shows the saturated physics-informed lane we avoid.
Fusion Equilibrium Challenge (arXiv:2609.01750)	2026	Flux-map reconstruction from non-magnetic diagnostics + cross-machine transfer; 9,121 DIII-D + 2,416 MAST shots (133 GB)	Open competition; EFIT-derived ground truth	Baseline SSIM 0.83 on DIII-D, collapses on MAST	Static equilibrium, per-shot. Real-data option; cross-machine generalization is an open frontier.

Literature Review – Comparison Table (E: Latent / Missing-Data Representations)

Paper (Year)		Problem / Dataset	Method	Key Result	Limitation / Relevance
Kit et al. (arXiv:2308.14556)	2023	Learn low-dim state representation + forward dynamics; AUG T_e/n_e profiles + machine parameters	VAE state representation + forward model	Predicts profile evolution from machine parameters; H-mode separable	Uses full profiles as input (not sparse sensors), discrete-time. Latent-state plasma learning precedent; we add sparse-sensor input + continuous time.
Yang et al., FusionMAE (arXiv:2509.12945)	2025	Unify 88 diagnostics into one embedding; virtual backup diagnosis; HL-3, 2,445 discharges, 88 channels	Masked autoencoder (transformer)	96.7% reliability on masked channels; emergent downstream gains	1D signals, masked-imputation (not dynamics). Missing-channel robustness precedent via masking (ours: via continuous time).

Research Gap

- All plasma **full-state reconstruction from sparse sensors** work uses **discrete-time** LSTM / Transformer encoders that assume regular, complete sampling.
- All **continuous-time latent dynamics** work (Neural ODE, LTC, CfC) is demonstrated on generic physics / medical / forecasting tasks – **not** on reconstructing plasma fields from sparse diagnostics.
- Real tokamak diagnostics are **irregularly sampled and dropout-prone** – exactly the condition continuous-time models are built for – but this robustness has never been exploited for plasma state reconstruction.

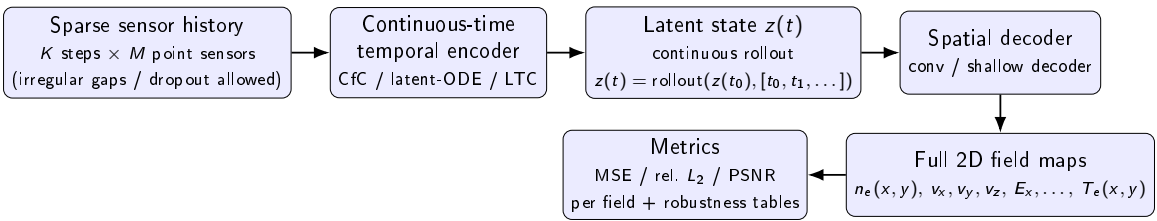
Gap, in one sentence

Sparse-diagnostic $\rightarrow$ full-state reconstruction of fusion plasmas has only been demonstrated with discrete-time encoders; continuous-time latent dynamics have never been brought to this task, nor to its irregular-sampling condition.

Project Objectives

1. Build a sensor $\rightarrow$ latent $\rightarrow$ full-field reconstruction model whose latent state evolves in **continuous time** (CfC / latent ODE).
2. **Match or beat** the published LSTM-encoder baseline (Reza-Faraji 2026 / plasma-SHRED 2025 setup) on clean, regularly-sampled data (same testbed, same 3 sensors).
3. **Demonstrate superiority** under irregular sampling and missing channels (dropout stress test), where discrete encoders degrade and the continuous-time model does not.
4. **Evaluate continuous-time roll-forward**: reconstruct at in-between / future query times beyond the observed window.
5. **Ablate** sensor count / placement and CfC-vs-LTC / latent-ODE variants; report full-field metrics.

Proposed Methodology – Architecture & Data Flow



Key differentiator vs. prior art: continuous-time evolution (not fixed discrete steps), robustness to irregular/missing data, and the ability to query the reconstruction at arbitrary times (roll-forward).

Proposed Methodology – Models & Loss Formulation

Temporal encoder. CfC (Closed-form Continuous-time; Hasani et al., 2022) is primary – it solves the continuous-time dynamics in closed form (no numerical ODE solver), giving speed and training stability. LTC (Hasani et al., 2021) and latent-ODE (Rubanova et al., 2019, built on Neural ODEs, Chen et al., 2018) are evaluated as contrasting variants.

The latent state evolves as a continuous-time ODE, integrated from the observation window into the query horizon:

$$\frac{dz(t)}{dt} = f_\theta\big(z(t), x(t)\big), \qquad z(t) = z(t_0) + \int_{t_0}^{t} f_\theta\big(z(\tau), x(\tau)\big)\, d\tau.$$

CfC replaces this integral with a closed-form expression, avoiding a numerical solver while retaining continuous-time semantics.

Spatial decoder. Convolutional / shallow decoder (SHRED-style), mapping the latent state to full-grid, multi-field output.

Reconstruction loss (summed over F fields, query times t):

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_{f=1}^{F} \big\| \hat{y}_f(t) - y_f(t) \big\|_2^2 \; \big(+ \; \lambda\, \mathcal{L}_{\text{back-proj}} \big)$$

$\mathcal{L}_{\text{back-proj}}$ (optional) re-projects predicted fields onto the sensor measurement operator.

Uniqueness / Novelty / Contribution

- **First application** of continuous-time latent dynamics to sparse-sensor → full-field **plasma state reconstruction** (prior art uses discrete-time LSTM encoders – Table A).
- Exploits continuous-time robustness to **irregular sampling and channel dropout** – directly matching the character of real fusion diagnostics (Table B), rather than assuming clean regular sampling.
- Enables reconstruction at **arbitrary query times** (interpolation and short extrapolation / roll-forward), which fixed-step discrete encoders cannot natively provide.
- Positioned as a **targeted architectural substitution** (LSTM → CfC) within an established reconstruction pipeline, isolating the contribution of continuous-time dynamics through controlled, like-for-like baselines (Objective 2) and stress tests (Objective 3).

This claim is consistent with the literature review: no reviewed paper combines continuous-time latent dynamics with sparse-sensor, full-field plasma reconstruction.

Dataset & Experimental Setup

Data source. [FILL IN: confirm testbed – e.g. 2D PIC E×B kinetic simulation, matching Reza & Faraji (2026) / SHRED-E×B (2025) setup, for direct comparability with the baseline.]

Sensor setup. Virtual point sensors (density, following the baseline's 3-sensor configuration); sensor count varied $\{3, 5, 10, 20\}$ for ablation (Objective 5).

Preprocessing. [FILL IN: normalization scheme, train/val/test split, handling of stiff dynamic ranges – see Limitations.] **Missing-data protocol.** Dropout $\{0\%, 20\%, 50\%\}$; irregular sampling injected for the stress-test (E2).

Baselines.

1. LSTM encoder + spatial decoder (reproduces Reza-Faraji 2026 / plasma-SHRED 2025) – main prior-art baseline
2. Static snapshot decoder (no temporal history)
3. GRU encoder (second discrete RNN)
4. Latent-ODE encoder (continuous-time, ODE-solver based – contrasts CfC)
5. DMD/SVD reduced-order reconstruction (classical linear baseline)

Environment. [FILL IN: hardware, framework/library versions.]

Implementation Status / Preliminary Results

[UPDATE BEFORE SUBMISSION – report actual progress]

- ☐ Simulator validated and generating snapshots
- ☐ Virtual sensor extraction
- ☐ CfC / LSTM models implemented
- ☐ Training runs + curves
- ☐ Preliminary reconstruction images

FIGURE PLACEHOLDER

training/validation curves, or

sample reconstruction vs. ground truth

Challenges / Limitations

- **Simulation-data realism** vs. real tokamak complexity – mitigated by using a validated benchmark testbed.
- **Extreme dynamic ranges / stiff dynamics** – mitigated via normalization and the closed-form CfC formulation.
- **Ill-posedness**: several fields may be under-determined from one measured quantity – an honest limitation; we report per-field performance rather than an aggregate score alone.
- **Compute cost** of high-fidelity simulation data – kept tractable by staying in 2D with modest grid sizes.
- **Sensor noise** – addressed by adding noise to virtual sensors during training/evaluation.

Work Plan & Timeline

Task	Status	Owner(s)
Literature review & gap finalization	[FILL IN]	[FILL IN]
Simulator setup & validation	[FILL IN]	[FILL IN]
Virtual sensor extraction pipeline	[FILL IN]	[FILL IN]
Baseline (LSTM/GRU/DMD) implementation	[FILL IN]	[FILL IN]
CfC / LTC / latent-ODE implementation	[FILL IN]	[FILL IN]
E1: clean-data comparison vs. baseline	[FILL IN]	[FILL IN]
E2: irregular sampling / dropout stress test	[FILL IN]	[FILL IN]
E3: roll-forward (interp. + extrapolation)	[FILL IN]	[FILL IN]
E4: sensor-count ablation	[FILL IN]	[FILL IN]
Final report & presentation	[FILL IN]	[FILL IN]

[FILL IN: mark each row completed / current / remaining, and confirm per-member duties.]

Expected Outcomes & Evaluation Plan

Metrics. Per-field MSE, relative L_2, PSNR (and SSIM on image outputs); correlation of physically derived quantities (e.g. energy).

Experiments.

- E1 Equal footing on clean, regular data (vs. LSTM baseline)
- E2 Irregular sampling / missing-channel stress test (robustness tables)
- E3 Roll-forward horizon (interpolation + extrapolation)
- E4 Sensor-ablation curves (sensor count / placement)
- E5 Preliminary: training/validation curves, sample reconstructions vs. ground truth

Expected outcome

On par with or better than the LSTM baseline on E1 (clean, regular data); **clearly better** on E2–E3 (irregular sampling, missing channels, and roll-forward), where the continuous-time formulation is expected to show its advantage.

References I

- Reza & Faraji, "ML-Enabled Full-State Reconstruction of Fusion Plasmas from Minimal Sensor Measurements," arXiv:2607.04390, 2026.
- "Shallow Recurrent Decoder for Reduced-Order Modeling of E×B Plasma Dynamics," MLST/IOP, doi:10.1088/2632-2153/adcd20, 2025.
- Williams, Zahn & Kutz, "Sensing with Shallow Recurrent Decoder Networks (SHRED)," 2023.
- Shousha et al., "ML-based real-time kinetic profile reconstruction in DIII-D (RTCAKENN)," Nucl. Fusion 64:026006, 2023.
- "Temporal super-resolution of kinetic plasma profiles at COMPASS," PPCF (ae72c6), 2026.
- "Physics-informed deep learning model for line-integral diagnostics (Onion)," Nucl. Fusion (ade0ce), 2025.
- Shen et al., "Density profile reconstruction with KAN on EAST (PIDP-KAN)," Plasma Sci. Technol. (adca8e), 2025.
- "TSER for reconstructing missing electron temperature (EAST)," Nucl. Fusion (addb5f), 2025.
- Ferreira, Carvalho & Fernandes, "Full-Pulse Tomographic Reconstruction with Deep Neural Networks," Fusion Sci. Technol. 74(1-2):47, 2018.
- "Machine learning enhanced tomographic reconstruction for multispectral imaging on TCV," PPCF (ada856), 2025.
- "Time-resolved, physics-informed NN (TRT-PINN) for tokamak total emission," Nucl. Fusion (adb3bc), 2025.
- Rutigliano et al., "Multi-diagnostics reconstruction of magnetic equilibrium and kinetic profiles using PINNs (JET)," Nucl. Fusion 66:046006, 2026.
- "Fusion Equilibrium Challenge," arXiv:2609.01750 (NeurIPS-2026 benchmark), 2026.
- Kit et al., "On learning latent dynamics of the AUG plasma state," arXiv:2308.14556, 2023.
- Yang et al., "FusionMAE: pretrained model for fusion diagnostics," arXiv:2509.12945, 2025.

References II

Method background (cited in text):

- Chen et al., "Neural Ordinary Differential Equations," NeurIPS, 2018.
- Rubanova, Chen & Duvenaud, "Latent ODEs for Irregularly-Sampled Time Series," NeurIPS, 2019.
- Hasani et al., "Liquid Time-constant Networks," AAAI, 2021.
- Hasani et al., "Closed-form Continuous-time Neural Networks (CfC)," Nature Machine Intelligence, 2022.